2008 ENGINE

Exhaust System - Edge & MKX

SPECIFICATIONS

TORQUE SPECIFICATIONS

Description	Nm	lb-ft
Bracket-to-catalytic converter bolts	20	15
Catalytic converter support bracket-to-engine block bolts	40	30
Catalytic converter support bracket-to-engine block nut	40	30
Catalytic converter support bracket-to-transmission bolts	48	35
Catalytic converter-to-exhaust manifold nuts	40	30
Catalytic converter-to-exhaust manifold studs	25	18
Exhaust flexible pipe-to-exhaust Y-pipe nuts	40	30
Exhaust Y-pipe-to-catalytic converter nuts	40	30
Power steering rack shield bolts	15	11
Roll restrictor bolt	90	66
Roll restrictor shield nuts	11	8
Torca® clamp	47	35
Universal joint (U-joint) flange bolts	70	52

DESCRIPTION AND OPERATION

EXHAUST SYSTEM

The exhaust system consists of:

- an exhaust flexible pipe.
- two muffler assemblies.
- one resonator assembly.
- muffler brackets with isolators bolted to the body.
- an exhaust Y-pipe.
- a catalyst monitor sensor mounted to each catalytic converter.
- a LH and RH catalytic converter.

The exhaust system provides an exit for exhaust gases and reduces engine noise by passing exhaust gases through the catalytic converters, a muffler assembly and resonator. Rubber exhaust hanger isolators attach the exhaust system to the mounting hooks.

Catalytic Converter

The catalytic converter plays a major role in the emission control system. The catalytic converter operates as a gas reactor. Its catalytic function is to speed the heat-producing chemical reaction of components in the exhaust gases in order to reduce air pollutants.

The catalyst material inside the catalytic converter consists of a ceramic substrate.

Precautions

CAUTION: Do not use leaded fuel in a vehicle equipped with a catalytic converter. In a vehicle that is continually misfueled, the lead in the fuel will be deposited in the catalytic converter and completely blanket the catalyst. Lead reacts with platinum to "poison" the catalyst. Continuous use of leaded fuel can destroy the catalyst and render the catalytic converter useless.

CAUTION: The addition of lead to the catalytic converter can also solidify the catalyst, causing excessive back pressure in the exhaust system and possibly causing engine damage.

CAUTION: Extremely high temperatures of 1,100°C (2,012°F) or above due to misfiring or an over-rich fuel/air mixture will cause the ceramic substrate to sinter or burn, destroying the catalytic converter. Do not continue to operate the vehicle if the engine is misfiring, there is a power loss, or other unusual operating conditions, such as engine overheating and backfiring.

The catalytic converter is designed to provide a long life. No maintenance is necessary for the catalytic converter.

Sound Insulators and Shields

Sound insulators and shields, attached to the underbody, protect the vehicle from exhaust system heat and should be inspected at regular intervals to make sure they are not dented or out of position. If a sound insulator and shield is damaged or shows evidence of deterioration, it should be replaced. The sound insulators and shields for the muffler, muffler pipe, resonator and catalytic converter pipe are installed separately.

DIAGNOSTIC TESTS

EXHAUST SYSTEM

Special Tools

Illustration	Tool Name	Tool Number
 ST2311-A	Electronic Vibration Analyzer or equivalent	100-F027 (014-00344)

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect the components of the exhaust system and related controls that may affect exhaust gas quality or loss of power.
3. Visually inspect for obvious signs of mechanical damage. Refer to the following chart.

VISUAL INSPECTION CHART

Mechanical
• Exhaust pipe pinched or crushed • Damaged muffler • Broken or damaged exhaust hanger brackets • Damaged catalytic converter • Cracked exhaust manifold • Loose or damaged heat shields

4. Verify that the exhaust system is installed correctly, with clamps correctly located and tightened to specification.
5. If the fault is not visually evident, determine the symptom. Go to **Symptom Chart - Exhaust System** or Go to **Symptom Chart - Noise, Vibration and Harshness (NVH)**.

Symptom Chart - Exhaust System

Symptom Chart - Exhaust System

Condition	Possible Sources	Action
• Vehicle has low or no power - vehicle performance complaint	• Exhaust pipe pinched or crushed • Damaged catalytic converter • Loose obstruction in exhaust	• INSPECT the exhaust components for damage. REPAIR or INSTALL new components as necessary. TEST the system for normal operation. If the concern is still present, REFER to the **Introduction - Gasoline Engines** article.
	• Restricted exhaust (possible frozen condensate in muffler)	• CHECK drain holes for debris. PARK the vehicle inside to thaw. TEST the vehicle for normal operation. If the concern is still present, REFER to the **Introduction - Gasoline Engines** article.

• Burning smell - usually occurs at idle, with possible traces of smoke	• Foreign material caught in exhaust system • Missing heat shields	• INSPECT the exhaust system for debris or missing heat shields. REPAIR or INSTALL new components as necessary. TEST the system for normal operation after the repair.
• Odor - described as a sulfur or rotten egg smell	• Catalytic converter • Excessive sulfur content in fuel	• At times, a slight sulfur smell is normal for catalytic converters. The cause is the sulfur content in the gasoline being used. ADVISE the customer no repair is required.
	• Rich fuel conditions • Misfire conditions	• REFER to the **Introduction - Gasoline Engines** article.
• Visible rust on surface of exhaust pipes	• Catalytic converter/exhaust system	• Surface rust is a characteristic of materials used on exhaust systems. Exposure to heat or road salt may result in surface rust. INSPECT for perforations. If there are no perforations, the condition is normal.

Symptom Chart - Noise, Vibration and Harshness (NVH)

NOTE: Noise, vibration and harshness (NVH) symptoms should be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to NOISE, VIBRATION AND HARSHNESS article. Since it is possible any one of multiple systems may be the cause of a symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to NOISE, VIBRATION AND HARSHNESS article for the next likely system and continue diagnosis.

Symptom Chart - Noise, Vibration and Harshness (NVH)

Condition	Possible Sources	Action
		• INSPECT the exhaust system for loose or missing heat shields or foreign material trapped between the heat

• Rattle, squeaks or buzz type noise - from the bottom of the vehicle	• Loose or damaged heat shield	shields and the exhaust system components. If any heat shields are loose, INSTALL worm gear clamp 7L5Z-5A231-AA and tighten to 7 Nm (62 lb-in). If the heat shields are missing, INSTALL new heat shields or exhaust system components as necessary. If a rattle, noise or buzz condition persists, INSTALL a new heat shield or component as necessary. TEST the system for normal operation after the repair.
	• Loose or damaged exhaust isolators	• VERIFY that the exhaust isolators are correctly installed. INSPECT the exhaust isolators for wear or damage. INSTALL new isolators as necessary. TEST the system for normal operation after the repair.
	• Damaged exhaust isolator hanger bracket	• INSPECT the exhaust system components for damage or broken hangers. INSTALL new components as necessary. CHECK for loose or damaged exhaust hanger brackets or fasteners. TIGHTEN the bolts to specification or INSTALL new components as necessary. TEST the system for normal operation after the repair.
	• Loose or damaged catalytic converter or muffler	• MOVE the exhaust system to simulate the bouncing action of the vehicle, checking for exhaust-to-body contact while moving the exhaust system. Using a rubber mallet, TAP on the exhaust components to duplicate the noise concern. Lightly TAP on the muffler, then the catalytic converter. DETERMINE if there are loose or broken baffles in the

		muffler or a loose or broken element in the catalytic converter. REPAIR or INSTALL new components as necessary. TEST the system for normal operation after the repair.
	• Exhaust grounded to chassis	• INSPECT for signs of exhaust components-to-body contact. If necessary, CARRY OUT the **Exhaust System Alignment**.
• Drone or clunk type noise - from the bottom of the vehicle	• Loose or damaged exhaust isolators	• INSPECT the exhaust isolators for wear or damage. INSTALL new isolators as necessary. TEST the system for normal operation after the repair.
	• Exhaust grounded to chassis	• INSPECT for signs of exhaust components-to-body contact. If necessary, CARRY OUT the **Exhaust System Alignment**.
• Whistles, boom, hum or ticking type noise - noise tends to change as the engine warms. The noises are often accompanied by exhaust fumes	• Exhaust system leak	• INSPECT the entire exhaust system for leaks. CHECK for punctures, loose or damaged clamps/fasteners, gaskets, sensors or broken welds. EXAMINE the chassis for grayish-white or black exhaust soot, which indicates exhaust leakage at that point. To magnify a small leak, have an assistant hold a rag over the tailpipe outlet while listening for a leak. REPAIR or INSTALL new components as necessary. TEST the system for normal operation after the repair.
		• MOVE the exhaust system to simulate the bouncing action of the vehicle, checking for exhaust-to-body contact while moving the exhaust system.

	• Catalytic converter	Using a rubber mallet, TAP on the exhaust components to duplicate the noise concern. Lightly TAP on the muffler and the catalytic converter. DETERMINE if there are loose or broken baffles in the muffler, or a loose or broken element in the catalytic converter. REPAIR or INSTALL new components as necessary. TEST the system for normal operation after the repair.
	• Exhaust muffler/resonator drain hole enlarged due to corrosion	• CONFIRM the drain holes are the noise source. INSTALL new components as necessary. TEST the system for normal operation after the repair.
• Hissing or rushing noise - high frequency sound. Vehicle performance is unaffected	• Exhaust system. Exhaust flow through pipes	• CHECK the exhaust system for leaks. Using a rubber mallet, TAP on the exhaust components to duplicate the noise concern. Lightly TAP on the muffler and the catalytic converter. Determine if there are loose or broken baffles in the muffler, or a loose or broken element in the catalytic converter. REPAIR or INSTALL new components as necessary. TEST the system for normal operation after the repair.
• Pinging noise - occurs when exhaust system is hot, engine turned off	• Catalytic converter/exhaust system	• Cool down pinging is a result of the exhaust system expanding and contracting during heating and cooling. This is a normal condition.
• Vibration - occurs at idle and at low speeds. Also accompanied by a clunk or buzz type noise	• Loose or damaged exhaust isolator	• INSPECT the exhaust isolators for wear or damage. INSTALL new isolators as necessary. TEST the system for normal operation after the repair.

	• Loose or damaged exhaust isolator hanger brackets	• INSPECT the exhaust isolator hanger brackets for wear or damage. INSTALL or REPAIR as necessary. TEST the system for normal operation after the repair.
	• Damper broken or out of position	• CHECK for the correct damper orientation . RELOCATE to the correct position and tighten the nuts to specification. INSPECT for missing or damaged damper. INSTALL new components as necessary. TEST the system for normal operation after the repair.
	• Exhaust system grounded to chassis	• CARRY OUT the **Exhaust System Alignment**.
• Engine drumming noise - normally accompanied by vibration	• Damaged or misaligned exhaust system	• INSPECT the exhaust system for loose or damaged fasteners, Torca® clamps, or isolators. CARRY OUT the **Exhaust System Alignment**.
• Sputter type noise - noise worse when cold, lessens or disappears when the vehicle is at operating temperature	• Damaged or worn exhaust system	• INSPECT the exhaust system for leaks or damage. REPAIR as necessary. TEST the system for normal operation after the repair.
• Thumping noise - from the bottom of the vehicle, worse at acceleration	• Misaligned exhaust system	• CHECK the exhaust system to chassis clearance. CHECK the exhaust system isolators for damage. REPAIR as necessary. TEST the system for normal operation after the repair.
• Engine vibration - is felt with increases and decreases in engine RPM	• Strain on exhaust system isolators	• CARRY OUT the **Exhaust System Alignment**. REPAIR as necessary. TEST the system for normal operation after the repair.
• Drumming noise - occurs inside the vehicle during idle or high idle, hot or cold.	• Exhaust system vibration excites the body resonances	• Go to **Pinpoint Test A** .

Very low-frequency drumming is very RPM dependent	inducing interior noise	

Pinpoint Test

PINPOINT TEST A: DRUMMING NOISE

A1 CHECK THE EXHAUST SYSTEM

- Key in START position.
- Increase the engine RPM until the noise is the loudest. Note the engine RPM.
- Key in OFF position.

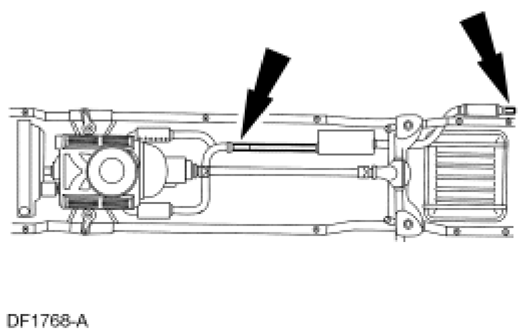

Fig. 1: Exhaust System
Courtesy of FORD MOTOR CO.

- Add approximately 9 kg (20 lb) of weight to the exhaust system. First place the weight at the tail pipe and test, then at the front pipe.
- Key in START position.
- Increase the engine RPM and listen for the drumming noise. Note the engine RPM if the noise occurs.
- Key in OFF position.
- Using a vibration analyzer (VA), determine the amount of vibration that occurs with the drumming noise.
- **Is the noise/vibration reduced or eliminated, or does the noise/vibration occur at a different RPM?**

  YES : REFER to **Exhaust System Alignment**. TEST the system for normal operation.

  NO : CONDUCT a diagnosis on other suspect systems. REFER to **NOISE, VIBRATION AND HARSHNESS** article.

GENERAL PROCEDURES

EXHAUST SYSTEM ALIGNMENT

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Loosen all fasteners joining the exhaust system components.
3. Beginning at the front of the vehicle, align the exhaust system to establish the maximum clearance. Make sure all fit pipes are pushed all the way into the preceding pipe and the notches are correctly lined up with the tabs.
4. Beginning at the front of the vehicle, tighten all fasteners and clamps to specification. For additional information, refer to **SPECIFICATIONS**.
5. Start the engine and check the exhaust system for leaks.

TORCA® CLAMP

1. Remove the nut from the Torca® clamp.
2. Grind the spot weld from the Torca® clamp and remove the clamp.
3. Clean the uneven surface area and position a Torca® clamp.

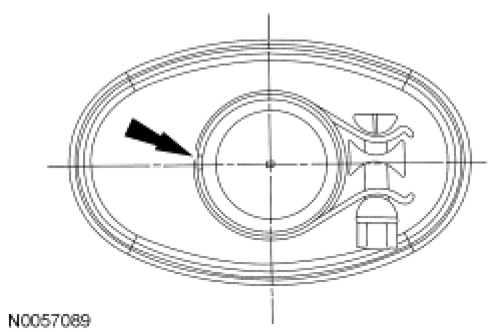

Fig. 2: Locating Uneven Surface Area
Courtesy of FORD MOTOR CO.

NOTE: **Make sure the clamp position is no more than 27.5 mm (1.08 in) or less than 25.5 mm (1 in) from the inlet of the resonator pipe.**

4. Make sure the back of the slot is covered by the clamp and the button is fully seated inside the notch.

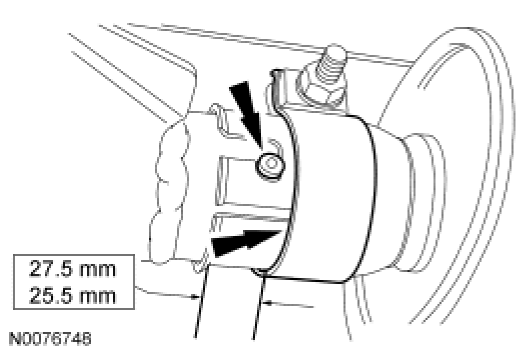

Fig. 3: Making Sure Back Of Slot Is Covered By Clamp & Button Is Fully Seated Inside Notch

With Specifications
Courtesy of FORD MOTOR CO.

NOTE: Do not tighten the clamp until the exhaust system has been aligned.

5. Tighten the Torca® clamp.
   - Tighten to 47 Nm (35 lb-ft).

REMOVAL AND INSTALLATION

EXHAUST SYSTEM - EXPLODED VIEW

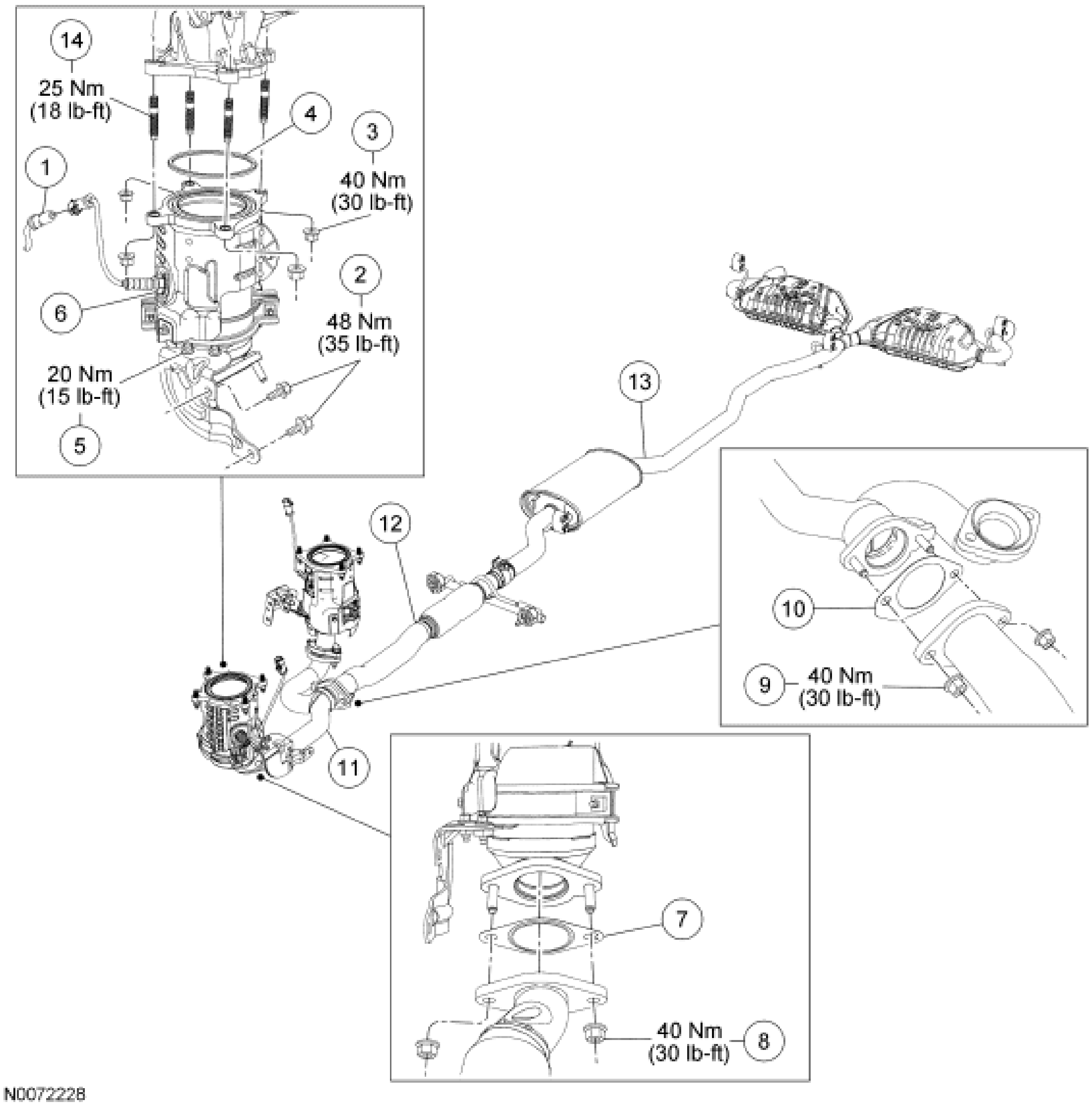

Fig. 4: Exploded View Of Exhaust System With Torque Specifications (1 Of 3)
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	14A464	LH catalyst monitor sensor electrical connector
2	W710656	LH catalytic converter support bracket-to-transmission bolts (2 required)
3	W705443	LH catalytic converter-to-exhaust manifold nut (4 required)
4	5F263	Gasket
5	5G228	Bracket-to-LH catalytic converter bolt (2 required)
6	5E213	LH catalytic converter
7	9451	Gasket
8	W705443	Exhaust Y-pipe-to-LH catalytic converter nut
9	W705443	Exhaust flexible pipe-to-exhaust Y-pipe nut (2 required)
10	9451	Gasket
11	5G274	Exhaust Y-pipe
12	5G203	Exhaust flexible pipe
13	5G213	Muffler and tailpipe
14	W712458	LH catalytic converter-to-exhaust manifold stud (4 required)

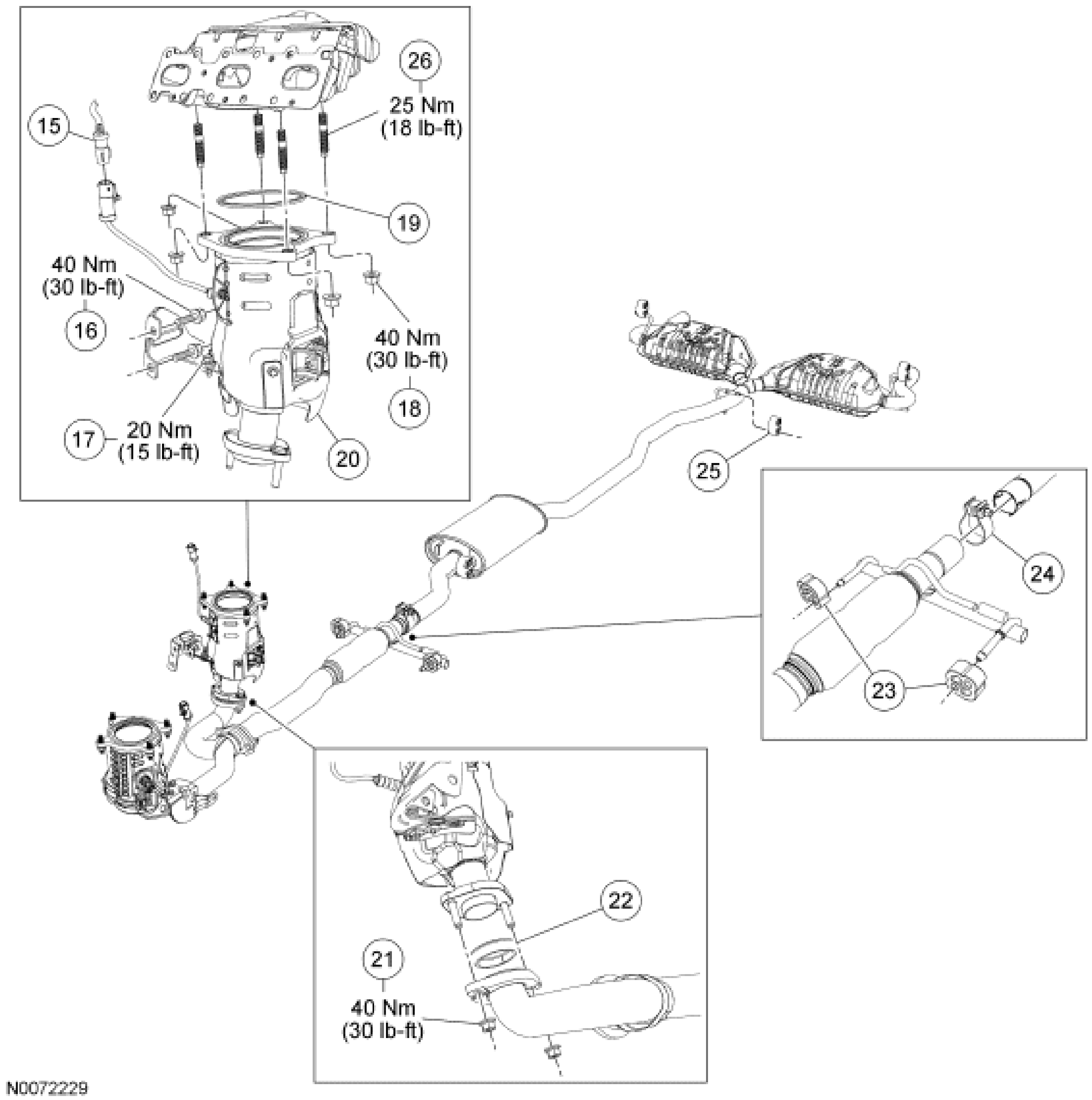

Fig. 5: Exploded View Of Exhaust System With Torque Specifications (2 Of 3)
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
15	14A464	RH catalyst monitor sensor electrical connector
16	W710512/ W705443	RH catalytic converter support bracket-to-engine block bolt (2 required) (front wheel drive [FWD] nut)
17	5G228	Bracket-to-RH catalytic converter bolt (2 required)
18	W705443	RH catalytic converter-to-exhaust manifold nut (4 required)
19	5F263	Gasket

20	5E211	RH catalytic converter
21	W705443	Exhaust Y-pipe-to-RH catalytic converter nut (2 required)
22	5E241	Gasket
23	5F262	Exhaust flexible pipe isolators (2 required)
24	5221	Torca® clamp
25	5F262	Muffler and tailpipe isolator (4 required)
26	W712458	RH catalytic converter-to-exhaust manifold stud (4 required)

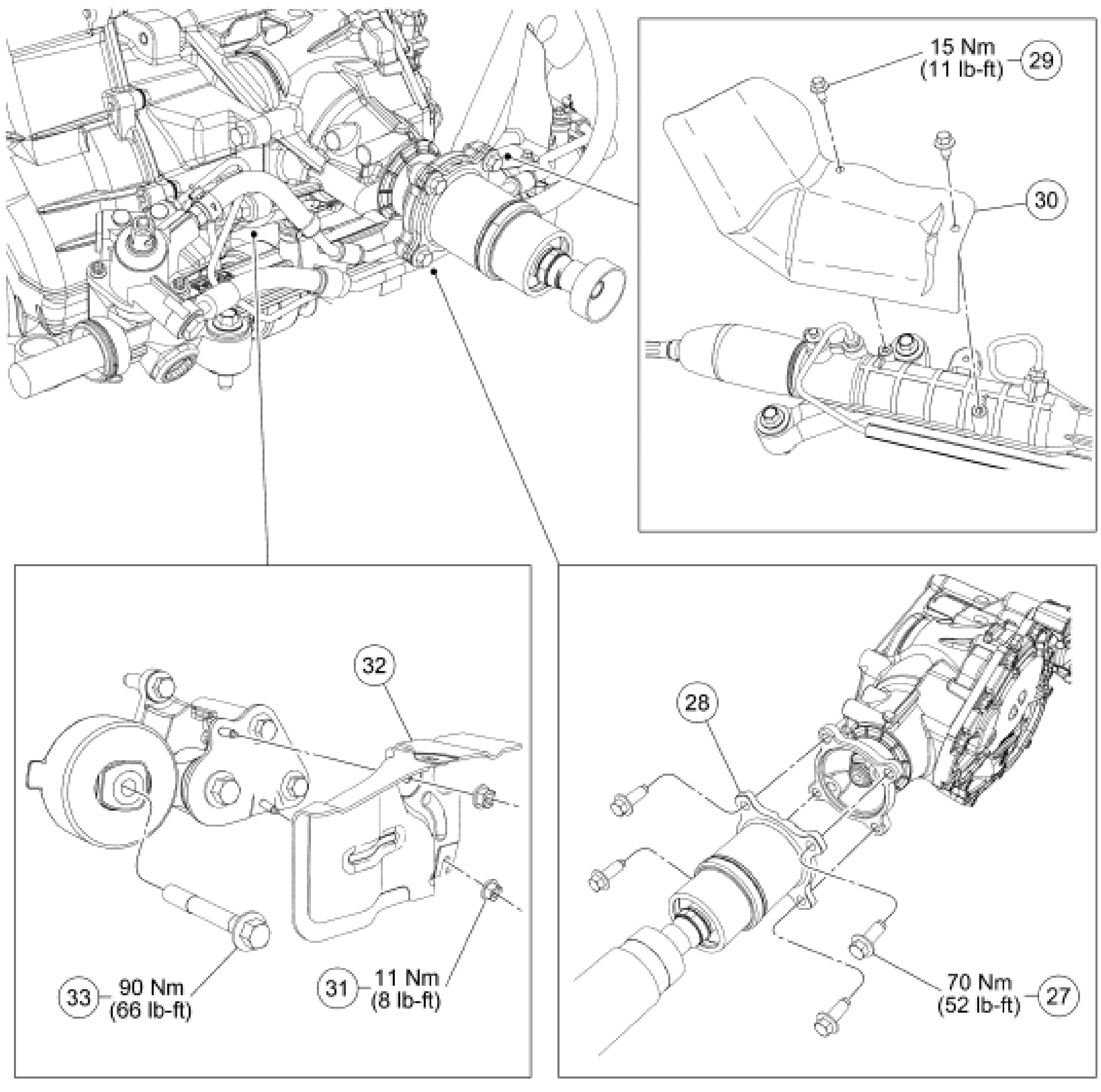

Fig. 6: Exploded View Of Exhaust System With Torque Specifications (3 Of 3)
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
27	W711918	U-joint flange bolts (4 required) (all wheel drive [AWD] only)
28	4K357	Front driveshaft (AWD only)
29	W500210	Power steering rack shield bolts (2 required)
30	-	Power steering rack shield
31	W700101	Roll restrictor shield nut (2 required)
32	6C038	Roll restrictor shield
33	W706674	Roll restrictor bolt

1. For additional information, refer to the appropriate procedure.

CATALYTIC CONVERTER - LH

REMOVAL AND INSTALLATION

NOTE: Always install new fasteners and gaskets. Clean flange faces prior to new gasket installation to make sure of correct sealing.

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Disconnect the catalyst monitor sensor electrical connector.
3. Remove the exhaust Y-pipe. For additional information, refer to **Exhaust Y-Pipe**.
4. Remove the 2 catalytic converter support bracket-to-transmission bolts.
   - To install, tighten to 48 Nm (35 lb-ft).
5. Remove the 4 nuts and the LH catalytic converter.
   - Discard the nuts and gasket.
   - To install, tighten to 40 Nm (30 lb-ft).
6. To install, reverse the removal procedure.
   - Install a new gasket and nuts.

CATALYTIC CONVERTER - RH

All vehicles

NOTE: If necessary, the catalytic converter heat shield can be serviced separately.

NOTE: Always install new fasteners and gaskets. Clean flange faces prior to new gasket installation to make sure of correct sealing.

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Remove the catalyst monitor sensor. For additional information, refer to **ELECTRONIC ENGINE CONTROLS** article.
3. Remove the exhaust Y-pipe. For additional information, refer to **Exhaust Y-Pipe**.

All wheel drive (AWD) vehicles

NOTE: Index-mark the driveshaft for installation.

4. Remove and discard the 4 U-joint flange bolts and separate the front driveshaft and secure it with a length of mechanic's wire.
   - To install, tighten to 70 Nm (52 lb-ft).
5. Remove the intermediate shaft. For additional information, refer to **FRONT DRIVE HALFSHAFTS** article.
6. Remove the 2 catalytic converter support bracket-to-engine block bolts.
   - To install, tighten to 40 Nm (30 lb-ft).

All vehicles

7. Remove the 2 bolts and the power steering rack shield.
   - To install, tighten to 15 Nm (11 lb-ft).

Front wheel drive (FWD) vehicles

8. Remove the catalytic converter support bracket-to-engine block bolt and nut.
   - To install, tighten to 40 Nm (30 lb-ft).

All vehicles

9. Remove the 2 nuts and the roll restrictor shield.
   - To install, tighten to 11 Nm (8 lb-ft).
10. Remove the roll restrictor bolt and rotate the engine forward.
    - To install, tighten to 90 Nm (66 lb-ft).

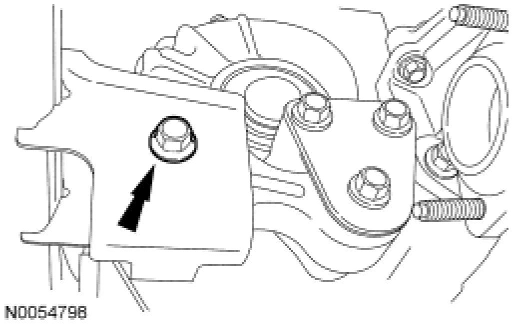

Fig. 7: Locating Roll Restrictor Bolt
Courtesy of FORD MOTOR CO.

11. Remove the 2 bracket-to-RH catalytic converter bolts.
    - To install, tighten to 20 Nm (15 lb-ft).
12. Remove the 4 nuts and the RH catalytic converter.
    - Discard the 4 RH catalytic converter nuts and gasket.
    - To install, tighten to 40 Nm (30 lb-ft).
13. To install, reverse the removal procedure.
    - Install a new gasket, nuts and studs.

EXHAUST FLEXIBLE PIPE

REMOVAL AND INSTALLATION

CAUTION: Do not excessively bend, twist or allow the exhaust to hang from the flexible pipe or damage to the exhaust system may occur.

NOTE: Always install new fasteners and gaskets. Clean flange faces prior to new gasket installation to make sure of correct sealing.

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Remove and discard the 2 exhaust flexible pipe-to-exhaust Y-pipe nuts.
    - Discard the gasket.
    - To install, tighten to 40 Nm (30 lb-ft).
3. Remove the Torca® clamp. For additional information, refer to **Torca® Clamp**.
4. Separate the 2 exhaust flexible pipe isolators and remove the exhaust flexible pipe.
5. To install, reverse the removal procedure.
    - Install a new gasket and nuts.
6. Start the engine and check for exhaust leaks.

EXHAUST Y-PIPE

REMOVAL AND INSTALLATION

CAUTION: Do not excessively bend, twist or allow the exhaust to hang from the flexible pipe or damage to the exhaust system may occur.

NOTE: Always install new fasteners and gaskets. Clean flange faces prior to new gasket installation to make sure of correct sealing.

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Remove and discard the 4 exhaust Y-pipe-to-LH and RH catalytic converter nuts.
    - Discard the gaskets.
    - To install, tighten to 40 Nm (30 lb-ft).
3. Remove and discard the exhaust Y-pipe-to-exhaust flexible pipe nuts and remove the exhaust Y-pipe.
    - Discard the gasket.
    - To install, tighten to 40 Nm (30 lb-ft).
4. To install, reverse the removal procedure.
    - Install new gaskets and nuts.
5. Start the engine and check for exhaust leaks.

MUFFLER AND TAILPIPE

REMOVAL AND INSTALLATION

NOTE: Do not use oil or grease-based lubricants on the isolators. They may cause deterioration of the rubber.

NOTE: Oil or grease-based lubricants on the isolators may cause the exhaust hanger isolator to separate from the exhaust hanger bracket during vehicle operation.

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Support the muffler and tailpipe with a suitable jackstand.

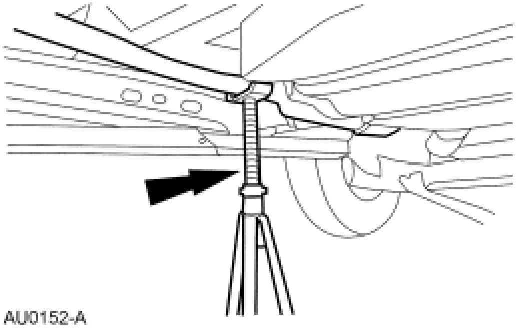

Fig. 8: Supporting Muffler & Tailpipe With A Suitable Jackstand
Courtesy of FORD MOTOR CO.

3. Remove the Torca® clamp. For additional information, refer to **Torca® Clamp**.

NOTE: Do not damage or tear the isolators during removal.

4. Using soapy water, separate the 4 muffler and tailpipe isolators from the vehicle.
5. Separate the muffler and tailpipe from the exhaust flexible pipe and remove the muffler and tailpipe.
6. To install, reverse the removal procedure.
   - Inspect and replace any isolators damaged or torn during the removal process.
7. Start the engine and check for exhaust leaks.